

Q1: Registers in AVR Microcontroller Ports

In AVR microcontrollers, each port contains three main registers:

- **DDR (Data Direction Register):** Determines whether the pin works as an input or an output.
- **PORT:** Used to write or send output values to the pin.
- **PIN:** Used to read the input value from the pin.

For the different ports, the registers are:

- PORTA → DDRA
- PORTB → DDRB
- PORTC → DDRC
- PORTD → DDRD

Q2: Difference Between SRAM and Flash Memory

SRAM:

- A volatile type of memory (data is lost when power is turned off).
- Stores variables while the program is running.
- Has faster access speed.

Flash Memory:

- A non-volatile type of memory (data remains even when power is off).
- Stores the program or code of the microcontroller.
- Access speed is slower compared to SRAM.